OCP GLOBAL SUMMIT 2026

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Open, Modular Rack-Scale AI Infrastructure for the Agentic Era

Dimitris Ziakas, CTO & VP Architecture

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The open, validated rack is now the center of AI infrastructure for the agentic era

Mix the best of the best. Lock into none of it.

ARCHITECTURE LENS

OCP-first

Open interfaces are the starting point, not the fallback.

SILICON LENS

Multi-vendor

Best-of-breed CPUs, accelerators, memory and fabric. No single roadmap.

SYSTEM LENS

Rack to POD

Compute, fabric and context memory, composed and validated as one system.

THE NEW QUESTION Which rack delivers the most useful tokens per watt, dollar and square foot?

Integrated, modular and open: the rack is now the engine of AI.

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Agentic AI moved the bottleneck off the accelerator

Three signals. One direction.

7x

year-over-year growth in monthly tokens processed across Google's surfaces

GOOGLE I/O 2026

4x+

more CPU capacity needed per gigawatt, inside the same power envelope

ARM

1:1–1:2

where CPU-to-GPU ratios are heading for agentic AI from 1:4–1:8 today

TRENDFORCE

Work around the model: Planning · Retrieval · Tool calls · Memory · Policy

The work around the model is now most of the work.

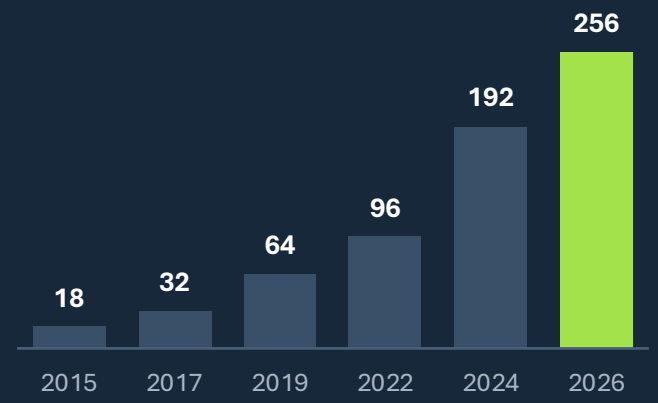
Sources: Google I/O 2026 keynote (blog.google); Arm AGI CPU launch (newsroom.arm.com); TrendForce, agentic AI and the CPU:GPU ratio (insights.trendforce.com). Third-party figures and projections, not Hyve forecasts.

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Agentic AI's three pillars, rebuilt over a decade

01 Compute 14x

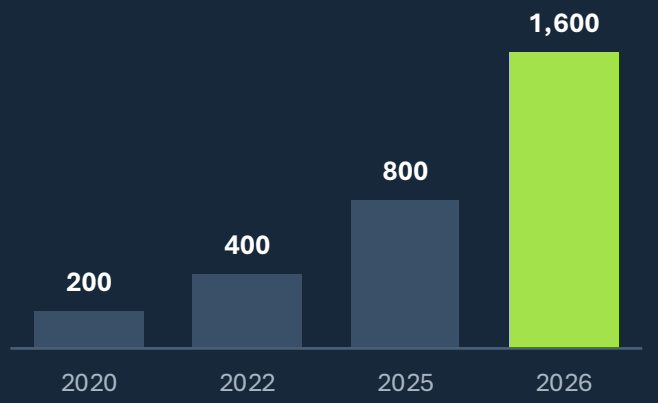
CPU orchestration + accelerator execution



Top server CPU cores per socket

02 Networking 8x

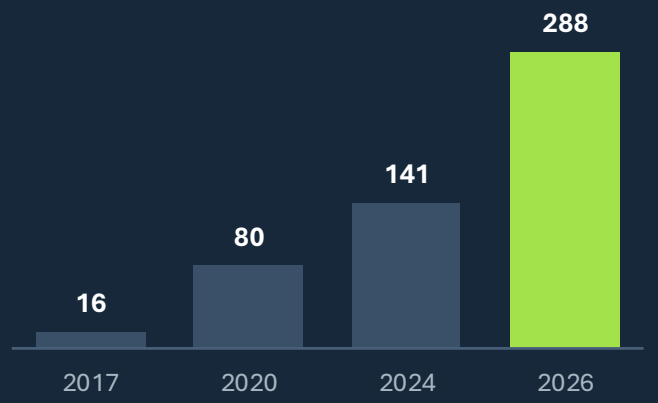
Scale-up + scale-out.
The fabric is the system bus.



Scale-out bandwidth per GPU (Gb/s)

03 Memory & storage 18x

HBM + DDR/CXL + NVMe.
Placement is the TCO lever.



HBM capacity per accelerator (GB)

Balance all three for your workload - or pay for capacity you never use.

Sources: Compute — Intel Xeon E5-2699 v3 (18 cores); AMD EPYC 7601/7742/9654/9965; AMD 6th Gen EPYC "Venice" (256 cores, launched July 2026). Networking — NVIDIA ConnectX-6/-7/-8 datasheets; ConnectX-9 in Vera Rubin NVL72 (1.6 Tb/s per GPU). Memory — NVIDIA V100 16GB, A100 80GB, H200 141GB, Rubin 288GB HBM4. Top-bin parts shown; year = general availability.

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East-west traffic now runs the cluster

Training and agentic inference flip the mix: most bytes now move GPU-to-GPU.

ALREADY TRUE IN THE CLOUD ERA

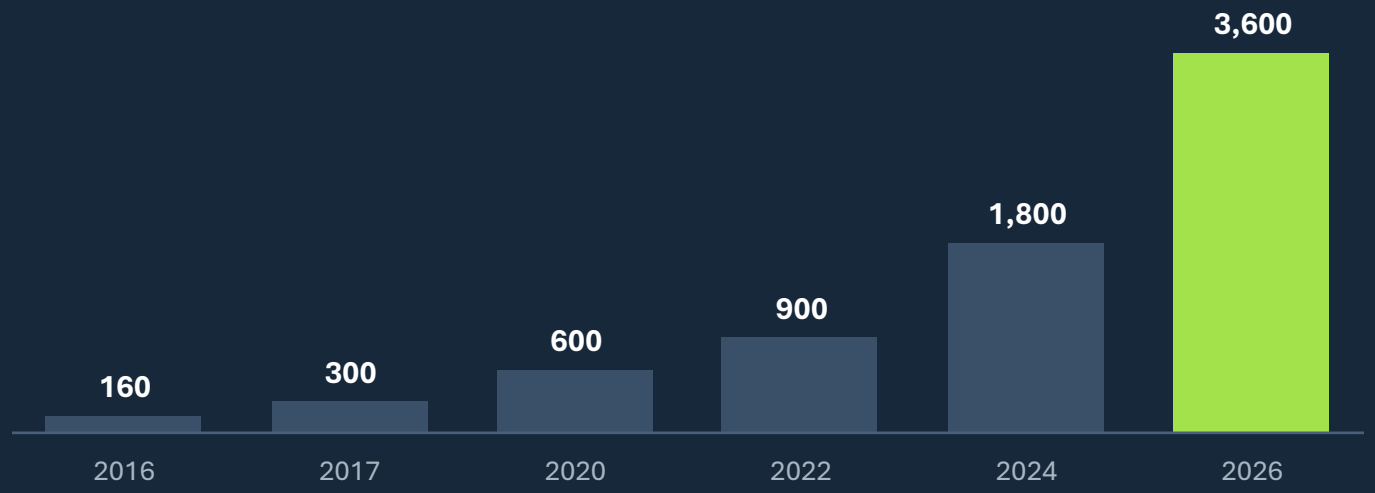
~3/4

of data center traffic already stayed inside the data center

What changed: the endpoints became GPUs and the bandwidth between them exploded.

SCALE-UP (GPU-TO-GPU) BANDWIDTH PER GPU, GB/s

22x



It’s all about the fabric, not the uplink.

Sources: Cisco Global Cloud Index (74.5% of data center traffic within the data center, 2017 forecast; >70% through 2021). NVLink 1–4 per-GPU bandwidth: WikiChip; NVLink 5 (Blackwell, 1.8 TB/s) and NVLink 6 (Vera Rubin NVL72, 3.6 TB/s): NVIDIA.

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Cabling and control are now architecture

CABLING & OPTICS

18x more fiber in a 72-GPU AI rack than a legacy CPU rack (576 fibers)

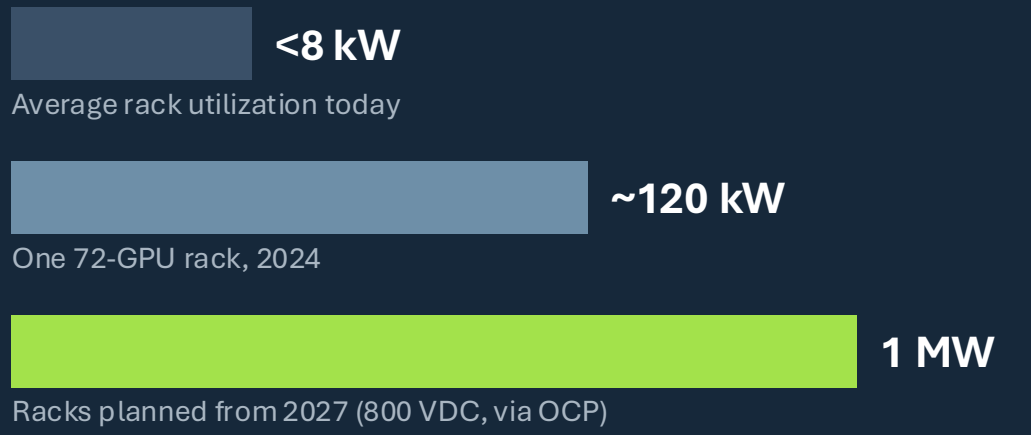
~2 mi of copper scale-up cabling — about 5,000 cables — inside one 72-GPU rack

10x+ more fiber in AI data centers than in traditional data centers

TELEMETRY & CONTROL

125x

Power per rack (log scale)



Power, liquid, fabric health, firmware: each now needs rack-level visibility and control.

Choice with discipline: open enough to avoid lock-in, engineered enough to deploy.

Sources: Corning (72 GPUs × 8 fibers = 576, 18x a legacy CPU rack; AI needs 10x+ more fiber); NVIDIA GTC 2024 via The Register and SemiAnalysis (GB200 NVL72: ~5,000 NVLink cables, ~2 miles, ~120 kW); Uptime Institute Global Data Center Survey 2024 (average rack <8 kW); NVIDIA 800 VDC architecture (1 MW racks from 2027).

KV Cache - Context and batching multiply memory, simultaneously

$$KV \approx 2 \times layers \times KV\text{-}heads \times head\text{-}dim \times tokens \times batch$$



What Breaks

At 128K context, two users fill an illustrative 80GB HBM budget; additional batching must shard or spill

What Fixes It

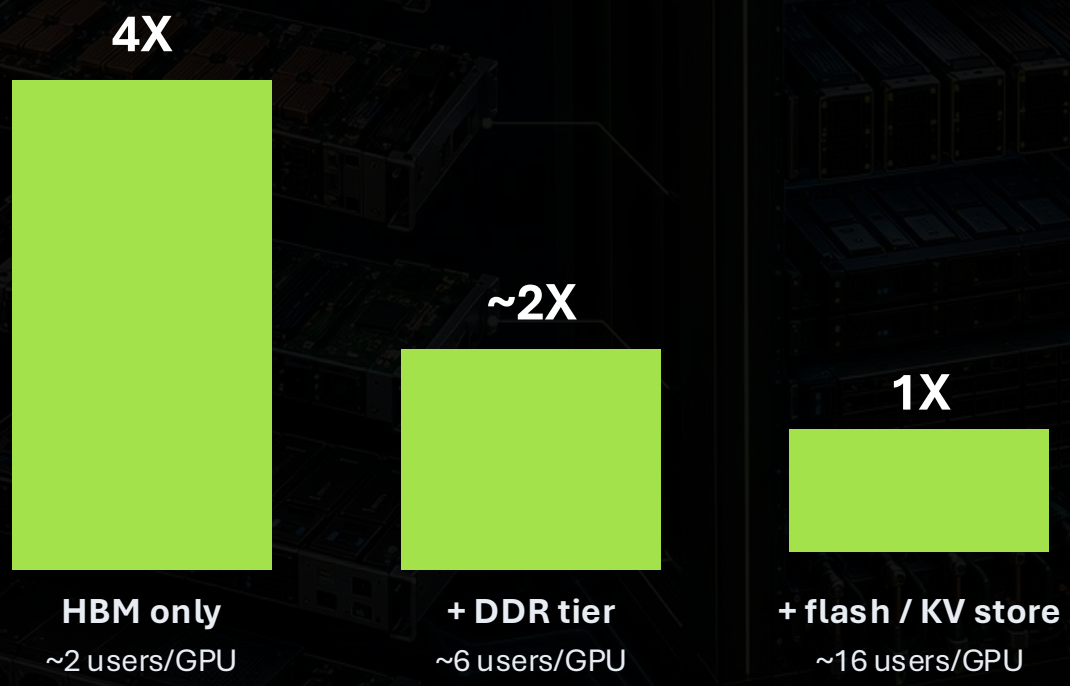
A tiered context architecture lets batching increase served users rather than forcing HBM-only scale-out.

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Tiered context memory is a utilization strategy, not cheaper storage

Same accelerator budget, more resident sessions, lower relative cost per served user.

RELATIVE COST PER SERVED USER



TAKEAWAY

The architecture decision is economic.

Allocate HBM to hot state. Use cheaper tiers for persistent context and batching.

Illustrative relative cost per served user (flash / KV-store tier = 1X). Not a Hyve price or performance guarantee.

For fixed long-context throughput, tiering changes the TCO curve

The biggest savings come from not buying accelerators just to hold context.

~65%

lower normalized 3-year TCO in this illustrative model

TAKEAWAY:

Allocate HBM to hot state and accelerator math; use cheaper tiers for persistent context and batching.

Illustrative normalized index (100 → 35). Not a Hyve price or performance guarantee. Replace with approved BOM/power model for any public dollar claim.

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Hyve makes open architecture deployable

Open, composable racks built from the best of the best, validated before they ship.

1 CHOOSE

- CPU ✓
- Accelerator ✓
- Memory & storage ✓
- Network fabric ✓
- Power & cooling ✓

Any vendor. No lock-in.

2 COMPOSE

- Network fabric
- Compute
- Compute
- Memory & storage
- Accelerators
- Accelerators
- Power & cooling

Open, modular, standards-based

3 DEPLOY

- Validated in advance
- Built around your workload
- Factory-integrated, ready to run

Engineer the choice. Validate the system. Deliver at speed.

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Agentic AI will reward the teams that design the whole system.

Compute, networking and context memory cannot be optimized independently.
Open, modular rack-scale architecture is the only scalable way to balance them.

That is where system architecture becomes a competitive advantage and it's where Hyve helps customers build for what comes next.

CONTINUE THE CONVERSATION

Visit Hyve Solutions at booth #C-31

Thank you.

Continue the conversation at the Hyve Solutions booth, #C-31.

Dimitrios Ziakas, CTO & VP Architecture

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Four standardization trends shaping rack-scale AI

Where the ecosystem is already converging and what it adds up to.

1

Open fabrics, scale-up and scale-out

GPU-to-GPU and node-to-node on open specifications.

UALink · UEC · ESUN

2

Tiered memory and context placement

A common way to describe HBM, DDR/CXL, NVMe — and where context lives.

CXL · KV-cache tiering

3

Rack power, cooling and telemetry

Megawatt-class racks need consistent, rack-level visibility and control.

800 VDC · liquid cooling · rack telemetry

4

Validation and deployment evidence

Qualified configurations, with evidence that travels with the rack.

Factory-to-site readiness

This is how OCP moves from component openness to rack-scale deployment confidence.

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KV cache turns storage into active AI infrastructure

Context placement is now a first-order TCO decision.

HBM
Hot state

›

DDR / CXL
Warm active

›

NVMe / Flash
Warm spill / reuse

›

Object / HDD
Cold storage

WHY IT MATTERS

Memory, not compute, caps users per accelerator.

THE ARCHITECTURE MOVE

Design server, storage, fabric and rack integration as one context-memory path.

THE HYVE MOVE

Right context, right tier. Don't buy accelerators to hold context.